

Numerical Analysis Assignment 3

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1 Problem 3.2

```

assignment 3/problem_3_2.py"
Values from least square: [ 2.96   1.746 -1.46   1.314]
Residual value: [0.00364]
Rank value: 4

```

Figure 1: Problem 3.2

From the Results of Problem 3.2, we can see that the values that we got are close to the direct values, with a little difference. Because we used more data, it is expected that the values gotten from least squares are more accurate than direct values. Also, as the residual value is small, we can conclude that the data that we have in our hand is accurate.

2 3.4

```

Solution for part (a): [1. 1.]
Solution for part (b): [ 7.00888731 -8.39566299]

Difference between the solutions: [-6.00888731  9.
39566299]

```

Figure 2: Problem 3.4 results.png

From the results, we can see that the difference between two solutions are big. The reason for this is that the matrix A has a big condition number:

```

Difference between the solutions: [-6.00888731  9.
39566299]
Condition number of A 1097.5386765221676
(base) Nijats-MacBook-Pro:assignment 3 nijatabbaso

```

Figure 3: Condition number of A

Because of the big condition number, the relative difference in the input results in big relative difference in the output.